

B8 v1 review checkpoint: cross-input coincidence flagging

Ticket (experimental-strategist, 2026-09-14, relaying Aaron's partial fleet-freeze lift): use simultaneous flags across independent inputs (box-air/box-gnd/viv-N/viv-E) as corroborating evidence for external RFI, to shrink the 68%-unknown bucket left by `flags@v0` (`marjum-2026-07/flags/v0/` , branch `rfi-flags-v0` , commit `824b0dd` , comb attribution corrected at `f4d06a3`). Validate on the same labelled windows used for v0: the 1.25 MHz/Panda comb event, the fan, and LIDAR. Report before/after unknown-fraction and category breakdown.

Question. Does "flag coincides across independent inputs at the same (time, channel)" separate external RFI from self-generated/instrumental RFI in this dataset, well enough to safely reclassify `unknown` pixels?

Inputs.

- `marjum-2026-07/flagging/{detectors,build_masks}.py` (v0, unmodified; re-run here on a handful of files, not the full campaign).
- `marjum-2026-07/curation/mode_table.jsonl` for live-input configuration and TX-comb on/off state per file.
- Raw data: `marjum-2026-07/data/*.h5` (11 GB, gitignored; read from the main checkout, not this worktree, since worktrees don't carry untracked data).
- Labelled windows from `CAMPAIGN.md` field notes (fan RFI 07-13, the 07-16 "RF resonance"/comb event tied to a Panda power-cycle test, the 07-17→07-18 digital self-comb, the 07-18 LIDAR sweep).

Units / encoding. uint8 category bitfield per (time, channel) sample, as defined in `flags/v0/flag_bits.json` ; channel width 0.244140625 MHz (250/1024). Bits: `cal`=1, `tx_comb`=2, `self-RFI`=4, `FM-scatter`=8, `airplane`=16, `orbcomm`=32, `unknown`=64, `overflow`=128. `cal/tx_comb/overflow` are non-RFI.

Assumptions going in (stated before running anything, so they can be checked rather than silently relied on):

1. The four named inputs are simultaneously live during the labelled windows, giving a genuine multi-way cross-check.

2. Different inputs are independent with respect to RFI, so coincidence is informative about external vs. self-generated origin.

Cheapest useful test. Re-run the existing (unmodified) v0 per-input detector on the small number of files spanning each labelled window (96 files total across 4 windows, seconds of compute -- not a campaign-wide rerun), and measure raw cross-input pixel coincidence against categories v0 already assigns on independent grounds (comb spacing, field-note power-cycle tests), rather than committing to a full re-flagging pipeline first.

Acceptance criteria for proceeding to a full v1 implementation: known self-generated events (comb-attributed `self-RFI`) should show low cross-input coincidence, and the unknown-fraction reduction should not come at the cost of relabelling known self-RFI as external.

Step 0 — topology check

Before measuring anything, confirm how many inputs are actually live at once during each labelled window.

```
In [1]: import json, sys, os
        from datetime import datetime, timezone
        # Code (this branch's own flagging/ modules) is resolved relative to this
        # notebook's own location, since worktrees mean the checkout containing
        # THIS branch's code may not be at any fixed absolute path.
        sys.path.insert(0, os.path.abspath(os.path.join(os.getcwd(), "..", "..", "flagging")))
        import build_masks as B
        import detectors as D

        # Default: this checkout's own marjum-2026-07/ (portable across machines --
        # Aaron's checkout is not at this server's path). Override only for a
        # worktree that doesn't physically have the gitignored raw data (worktrees
        # don't carry untracked files).
        MARJUM_ROOT = os.path.abspath(os.path.join(os.getcwd(), "..", ".."))
        DATA_ROOT = os.environ.get("MARJUM_DATA_ROOT", MARJUM_ROOT)
        modes = B.load_mode_table(os.path.join(DATA_ROOT, "curation", "mode_table.jsonl"))

        windows_ref = [
            ("07-13 fan window", "2026-07-13T05:05:00Z"),
            ("07-16 Panda event", "2026-07-16T01:05:00Z"),
```

```

    ("07-17/18 digital self-comb", "2026-07-18T00:30:00Z"),
    ("07-18 LIDAR sweep",          "2026-07-18T01:37:00Z"),
]
for label, t_str in windows_ref:
    t = datetime.strptime(t_str, "%Y-%m-%dT%H:%M:%SZ").replace(tzinfo=timezone.utc)
    target = f"corr_{t:%Y%m%d_%H%M%S}Z.h5"
    # mode_table rows can have gaps between contiguous-run boundaries, so
    # take the last row starting at or before the target rather than
    # requiring an exact bracket (which a mid-run timestamp can miss).
    cand = [m for m in modes if m["file_first"] <= target]
    m = cand[-1] if cand else None
    print(f"{label:30s} phase={m['phase']} if m else '?' } "
          f"live_inputs={m['live_inputs']} if m else '?' } tx_comb={m['tx_comb']} if m else
          '?' }")

```

07-13 fan window	phase=A	live_inputs=['3', '4']	tx_comb=off
07-16 Panda event	phase=C	live_inputs=['0', '4']	tx_comb=off
07-17/18 digital self-comb	phase=C	live_inputs=['0', '4']	tx_comb=off
07-18 LIDAR sweep	phase=C	live_inputs=['0', '4']	tx_comb=off

This confirms (and `flagging/coincidence_check_v1.py`'s own file-level scan re-confirms per-file) what `INDEX.md` already documents: **during Phase A the live pair is inputs 3,4 ; during Phase C — which covers all three of the 07-16/07-17/07-18 labelled windows — the live pair is 0,4 (box-gnd, box-air).** `viv-N / viv-E` are never simultaneously live with the box inputs in any of these windows. **Assumption 1 above is false for this campaign:** there is no window in which four independent inputs exist to cross-check against each other. The most this campaign ever offers is a 2-way check, and only between box-gnd and box-air, which share a platform, chassis, and power supply — the opposite of the geographically-independent-antennas picture the ticket's "coincidence implies external" reasoning assumes.

Step 1 — coincidence measurement on the labelled windows

Re-run `build_masks.process_file` (unmodified v0 code) on the files in each window, using the input pair that `mode_table.jsonl` says is actually live, and measure how often a pixel flagged in one input is also flagged (any RFI category) in the other input at the same (time, channel).

In [2]:

```

import importlib
sys.path.insert(0, os.path.abspath(os.path.join(os.getcwd(), "..", "..", "flagging")))
import coincidence_check_v1 as C
importlib.reload(C)
modes = C.mode_table()
results = [C.run_window(w, modes) for w in C.WINDOWS]
for r in results:
    print("---", r["window"])
    for k in ("n_files_used", "n_self_rfi_pixels_inputA",
              "frac_self_rfi_also_flagged_in_other_input",
              "n_unknown_pixels_inputA",
              "frac_unknown_also_flagged_in_other_input"):
        print(f"    {k}: {r.get(k)}")

```

```

--- 07-13 fan (~150 MHz, labelled self-RFI)
    n_files_used: 14
    n_self_rfi_pixels_inputA: 8631
    frac_self_rfi_also_flagged_in_other_input: 0.00011586142973004287
    n_unknown_pixels_inputA: 64087
    frac_unknown_also_flagged_in_other_input: 0.00010922652019910435
--- 07-16 Panda/1.25MHz-field-note event (labelled self-RFI, Panda power cycle)
    n_files_used: 13
    n_self_rfi_pixels_inputA: 0
    frac_self_rfi_also_flagged_in_other_input: None
    n_unknown_pixels_inputA: 54468
    frac_unknown_also_flagged_in_other_input: 0.7161636190056547
--- 07-17/18 digital self-comb, 1.953125 MHz (labelled self-RFI, conducted+radiated on BOTH live
inputs)
    n_files_used: 55
    n_self_rfi_pixels_inputA: 947315
    frac_self_rfi_also_flagged_in_other_input: 0.9524614304639956
    n_unknown_pixels_inputA: 403854
    frac_unknown_also_flagged_in_other_input: 0.7188934614984623
--- 07-18 LIDAR sweep, 15-min-interval RFI (labelled self-RFI, on-platform)
    n_files_used: 14
    n_self_rfi_pixels_inputA: 294044
    frac_self_rfi_also_flagged_in_other_input: 0.9289834174477289
    n_unknown_pixels_inputA: 125040
    frac_unknown_also_flagged_in_other_input: 0.8372600767754319

```

```

In [3]: import pandas as pd
rows = []
for r in results:
    rows.append({
        "window": r["window"][:42],
        "inputs": ", ".join(r["inputs"]),
        "n_files": r["n_files_used"],
        "self-RFI coincidence": r["frac_self_rfi_also_flagged_in_other_input"],
        "unknown coincidence": r["frac_unknown_also_flagged_in_other_input"],
        "any-RFI coincidence": r["frac_any_rfi_also_flagged_in_other_input"],
    })
df = pd.DataFrame(rows).set_index("window")
df

```

	inputs	n_files	self-RFI coincidence	unknown coincidence	any-RFI coincidence
window					
07-13 fan (~150 MHz, labelled self-RFI)	3,4	14	0.000116	0.000109	0.000126
07-16 Panda/ 1.25MHz-field- note event (labe	0,4	13	NaN	0.716164	0.760817
07-17/18 digital self-comb, 1.953125 MHz (	0,4	55	0.952461	0.718893	0.867941
07-18 LIDAR sweep, 15-min- interval RFI (la	0,4	14	0.928983	0.837260	0.898425

Findings

Window	Ground truth (independent evidence)	Self-RFI coincidence	Unknown coincidence
07-13 fan	self-generated (field notes; ~150 MHz band)	0.01%	0.01%
07-16 Panda event	self-generated (field notes: comb disappeared when the Panda was power-cycled)	n/a (0 self-RFI-attributed px on box-gnd)	71.6%
07-17→18 digital self-comb	self-generated (no beam/ pointing response; conducted on box-gnd's internal load and radiated on box-air — already documented in <code>INDEX.md</code> , corrected 2026-09-13)	95.2%	71.9%
07-18 LIDAR sweep	self-generated (on-platform pulsed source)	92.9%	83.7%

The naive rule is falsified by 3 of the 4 labelled windows. In every Phase-C window (07-16, 07-17/18, 07-18), pixels that other evidence independently confirms are self-generated show 70-95% cross-input coincidence — the opposite of what "coincidence implies external" predicts. This is not a surprising result in hindsight: box-gnd and box-air share a chassis and power supply, so a source that couples into one conducted path (a Panda EMI comb, a LIDAR pulse) routinely couples into the other at the same instant. Coincidence here measures shared platform environment, not external origin.

Only the 07-13 fan window shows near-zero coincidence, but that is not a confirmation of the rule either — it just means whatever produces the "self-RFI"/"unknown" flags on input 3 in Phase A does not reach input 4 at all in this sample; it does not establish that a genuinely external source would show low coincidence, since no unambiguously-external labelled window exists to test the positive-case direction. (The airplane and FM-scatter categories are morphological, not corroborated — the same caveat as v0's `MEMO.md` .)

Material uncertainties

1. **Topology mismatch.** The ticket's four-input premise doesn't hold in this campaign; the strongest cross-check ever available is 2-way, and the only 2-way pair actually exercised in the labelled windows (box-gnd/box-air) is platform-common, not independent.
2. **Sign is context-dependent, not fixed.** In the Phase-C windows, high coincidence tracks shared self-RFI, not corroborated externality — the inverse of the ticket's assumed direction. Any rule that reads "coincident $\Rightarrow$ more likely external" would, on this evidence, actively promote confidence in the wrong direction for Panda EMI, the digital self-comb, and LIDAR — three of the campaign's four best-documented self-RFI mechanisms.
3. **No confirmed-external labelled window exists** to test the other half of the hypothesis (would a real external signal actually show low coincidence here, e.g. because it's beam-limited or below one receiver's noise floor at a given moment?). Absence of that test case means I can't rule out coincidence being useful in the other direction either — this checkpoint only establishes that the rule as literally stated fails its self-RFI validation set.
4. This checkpoint used **only the pre-existing v0 detector**, re-run on 96 files (4 windows) out of 5,120 — a deliberately cheap smoke test, not a campaign-wide recomputation. No `flags/v0` output was touched.

Decision requested

Implementing "non-coincident unknown $\rightarrow$ self-RFI candidate, coincident unknown $\rightarrow$ likely external" as specified would misclassify the campaign's best-documented self-RFI mechanisms as more-likely-external. I have not implemented any reclassification or re-run the full campaign pending direction on how to proceed. Two options that would use the coincidence signal without its current failure mode:

- **(a) Restrict coincidence-based promotion to bands with no known self-RFI spectral signature** (i.e. exclude the comb bands and BAND_FAN/BAND_LAPTOP from any coincidence-driven reclassification), so it only touches the FM/DTV/orbcomm-adjacent `unknown` residue where self-RFI is not already suspected.
- **(b) Require an independent corroborating signature** (matched comb spacing, or the beam-response/no-pointing-dependence test already used for the digital self-comb) before trusting coincidence's sign at all, i.e. treat raw coincidence as too weak/ambiguous a signal on its own in this 2-input, platform-shared topology.

Awaiting Aaron's direction before scaling this to a full v1 pass.

STOPPED AT REVIEW GATE — awaiting Aaron's approval.